

# NICHOLAS ASHMORE

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Civil Engineer and Geology PhD passionate about addressing climate change through carbon dioxide removal innovation. Combines extensive academic and consultancy expertise with engineering mindset to solve complex problems. Highly proficient in programming (Python, MATLAB, C++) and geochemical modelling. A proven technical communicator, with experience securing grant funding, giving technical conference presentations, publishing scientific papers, teaching, and working within regulatory environments.

## Professional Experience

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### Hydrogeologist, Stantec, Edinburgh UK

2025 – Present

- Applied innovative analytical tools and numerical models to assess groundwater drought, surface water augmentation schemes and groundwater abstraction sustainability
- Communicated technical modeling outputs into actionable insights for water industry clients and environmental regulators, adapting presentation to audience technical knowledge
- Implemented team-wide code management practises using DevOps, improving code quality and increasing both collaboration and accessibility of Python code

### Subsurface CO<sub>2</sub> Storage Consultant, Part-time Contract, Isometric, London UK

2025 - Present

- Reviewed carbon dioxide removal suppliers CO<sub>2</sub> storage monitoring plans for compliance with local permitting requirements and scientific rigour
- Aligned existing CO<sub>2</sub> subsurface storage modules to streamline crediting process
- Prepared research documents on scientific developments relevant to business operations

### Doctor of Philosophy, Geology and Geophysics, University of Edinburgh

2021 – 2025

- Developed custom multiphase numerical models (MATLAB, C++) to simulate CO<sub>2</sub> migration impacts on aquifers and support monitoring design and data interpretation
- Worked as lead tutor supporting teaching of MSc and BSc in hydrogeology, environmental site assessment and numerical groundwater modelling.
- Presented findings at major industry-relevant conferences, and published findings in high-impact peer-reviewed journals

### Environmental Consultant, GSI Environmental, Austin TX

2020-2021

- Developed software package (DE-GAS Toolkit, available online) for characterization and interpretation of dissolved gas data using Visual Basic
- Monitored fugitive gas migration from leaking gas wells in the Marcellus shale through analysis dissolve gas and isotope data

### Master of Applied Science, Civil Engineering, Queen's University, Kingston, Canada

2018 - 2020

- Applied a reactive transport model (MIN3P) to simulate the dissolution and persistence of methane in the subsurface
- Designed and conducted laboratory experiments for numerical model validation

- Wrote successful grant application to secure funding from the Natural Sciences and Engineering Research Council of Canada
- Delivered tutorials as a teaching assistant in civil engineering students on subsurface contamination, numerical methods and groundwater engineering

**Undergraduate Research Assistant, Western University, London, Canada**

2016, 2017

- Designed and installed long-term field monitoring equipment (e.g., nested piezometers, multilevel pore-water samplers) to monitor geochemical processes
- Conducted long-term field monitoring of contaminant transport at groundwater – surface water interfaces on Great Lake beaches

**Conferences & Training**

Field Methods in Hydrogeology (University of Waterloo, 2019), Science Communication for the Geosciences Workshop (SAGES, 2024), Carbon Capture Utilisation and Storage Summer School (TU Delft, 2022)

Presentations delivered at InterPore (2025, 2023), EGU General Assembly (2024, 2026), SIAM Geosciences (2023)

**Education****Doctor of Philosophy, Geology and Geophysics, University of Edinburgh**

2025

**Master of Applied Science, Civil Engineering, Queen's University, Kingston, Canada**

2021

**Bachelor of Applied Science, Civil Engineering, Queen's University, Kingston, Canada**

2018

**Publications**

**Ashmore, N.A.**, Krol, M. M., Gilfillan, S. M. V., Van De Ven, C.J.C, Mumford, K. G., Molnar, I. L. (2024). Simulation of bench-scale CO<sub>2</sub> injection using a coupled continuum-discrete approach. *Science of the Total Environment*, 176639. <https://doi.org/10.1016/j.scitotenv.2024.176639>.

**Ashmore, N. A.**, Van De Ven, C. J. C, Mumford, K. G. (2024). Importance of Multicomponent Mass Transfer on the Dissolution of Trapped Gas: A Focus on Geo-Energy Applications in the Shallow Subsurface. *ACS ES&T Engineering*, 4(2), 290-299 <https://doi.org/10.1021/acsestengg.3c00150>.

**Ashmore, N. A.**, Van De Ven, C. J. C, Molofsky, L. J., & Mumford, K. G. (2022). Calculation of gas ratios for use in stray gas migration investigation and characterization. *Groundwater*, 60(3), 312-317, <https://doi.org/10.1111/gwat.13183>.

**Referees**

Dr. Ian Molnar, School of Geosciences, University of Edinburgh, [ian.molnar@ed.ac.uk](mailto:ian.molnar@ed.ac.uk)

Dr. Stuart Gilfillan, School of Geosciences, University of Edinburgh, [stuart.gilfillan@ed.ac.uk](mailto:stuart.gilfillan@ed.ac.uk)

Dr. Kevin Mumford, Department of Civil Engineering, Queen's University, [kevin.mumford@queensu.ca](mailto:kevin.mumford@queensu.ca)